

Lovely Professional University

School of Computer Science and Engineering UpGrad Campus, Phagwara,

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SYNOPSIS

for the B.Tech Final Year Project

Book Recommendation System

Submitted by:

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INTRODUCTION:

BACKGROUND/CONTEXT:

With the exponential growth of digital libraries and e-commerce platforms, readers are faced with an overwhelming number of book choices. Traditional browsing methods are time-consuming and often ineffective in helping readers discover books that truly match their preferences. Recommendation systems have become essential tools in the digital age, helping users navigate through vast collections and discover relevant content. A personalized book recommendation system can significantly enhance user experience by suggesting books aligned with individual reading preferences, genres, and rating patterns.

PROBLEM STATEMENT:

Readers struggle to discover new books that match their interests from the vast selection available in digital libraries and online bookstores. Traditional recommendation methods such as bestseller lists and editor picks do not account for individual preferences. Existing systems often lack accuracy in personalization and fail to consider multiple factors like user ratings, reading history, genre preferences, and book metadata. This project aims to develop an intelligent book recommendation system that utilizes collaborative filtering, content-based filtering, and hybrid approaches to provide accurate and personalized book suggestions to users.

RATIONALE/JUSTIFICATION:

Book recommendation systems have proven to increase user engagement and satisfaction in digital library platforms and e-commerce websites. This project addresses a real-world problem faced by online book retailers and libraries worldwide. By implementing various machine learning algorithms and recommendation techniques, we can demonstrate practical applications of data science in improving user experience. The project combines data preprocessing, feature engineering, machine learning, and web development—essential skills for modern software development.

OBJECTIVES:

GENERAL OBJECTIVE:

To design and implement an intelligent book recommendation system that utilizes machine learning algorithms to provide personalized book suggestions based on user preferences, ratings, and reading history.

SPECIFIC OBJECTIVES:

- Collect and preprocess book datasets from Kaggle containing book metadata, ratings, and user information.
- Perform exploratory data analysis (EDA) to understand book distribution, rating patterns, and user behavior. 11123 → rows & 12 → columns.
- Implement collaborative filtering algorithms (user-based, item-based, matrix factorization) to identify similar users and books.
- Develop content-based filtering using book features such as genre, author, publication year, and textual descriptions.
- Create a hybrid recommendation model combining collaborative and content-based approaches for improved accuracy.
- Build a React-based frontend interface for user interaction, profile management, and viewing recommendations.

- Develop a FastAPI backend to handle user requests, manage recommendations, and interact with the recommendation engine.

SCOPE OF THE PROJECT:

Included:

- Collection and preprocessing of book datasets from Kaggle (e.g., Goodreads, Amazon book datasets).
- Implementation of multiple recommendation algorithms (collaborative filtering, content-based, hybrid).
- User authentication and profile management system.
- Web-based interface for browsing, rating books, and receiving personalized recommendations.
- Performance evaluation using metrics like RMSE, MAE, precision, recall, and NDCG.
- Comparison of different recommendation algorithms and selection of the best-performing model.

Excluded:

- Real-time inventory management and book purchase functionality.
- Integration with external e-commerce platforms or payment gateways.
- Social features such as user reviews, comments, or social sharing.
- Mobile application development (focus on web application only).

Literature Review (brief):

- Collaborative Filtering: (People who liked similar things in the past will like similar things in the future) Goldberg et al. (1992) introduced the concept of recommendation systems using collaborative filtering to predict user preferences based on similar users' ratings.
- Content-Based Filtering (item itself) Lops et al. (2011) demonstrated how content-based approaches using item features can provide personalized recommendations without relying on user-to-user similarity.
- Matrix Factorization: Koren et al. (2009) (Find hidden patterns) showed that matrix factorization techniques significantly improve recommendation accuracy by discovering latent factors in user-item interactions.
- Hybrid Approaches: Burke (2002) (No single method is perfect, so combine them) demonstrated that hybrid recommendation systems combining multiple algorithms outperform individual approaches in accuracy and coverage.
- Deep Learning for Recommendations (Use neural networks to learn complex user behavior.) : He et al. (2017) introduced neural collaborative filtering, leveraging deep learning for improved recommendation performance.

METHODOLOGY:

Research Design:

Quantitative experimental approach using historical user rating data and benchmark evaluation of recommendation algorithms against standard metrics.

Data Collection Methods:

- Kaggle datasets: Book datasets containing book information, user ratings, and reading history.
- Data preprocessing: Handling missing values, removing duplicates, standardizing formats.

- Data splitting: Dividing into training, validation, and test sets (70-15-15 split).

Data Analysis Techniques:

- Exploratory Data Analysis: Statistical analysis of book distribution, rating distribution, and user behavior patterns.
- Feature Engineering: Extracting features from book metadata (genre, author, publication year, average rating).
- Algorithm Implementation: Collaborative filtering (user-based, item-based), matrix factorization (SVD), and hybrid approaches.
- Evaluation Metrics: RMSE, MAE, Precision@K, Recall@K, NDCG, Coverage, and Diversity.

Tools and Technologies:

- Languages: Python (ML/backend), JavaScript/React (frontend)
- ML Libraries: Scikit-learn, Pandas, NumPy, Surprise (recommendation library), TensorFlow/Keras
- Backend: FastAPI for REST API, SQLAlchemy for database ORM, PostgreSQL/SQLite for data storage
- Frontend: React, Axios for API calls, Chart.js/D3.js for data visualization
- Data Analysis: Jupyter Notebook, Matplotlib, Seaborn for visualization
- Deployment: Docker, AWS/Google Cloud or local server.

Expected Outcomes/Deliverables:

- A functional book recommendation system with both backend API and frontend web application.

- Trained recommendation models with performance metrics and comparative analysis.
- Comprehensive documentation including architecture, API specifications, user guides, and dataset details.
- Exploratory data analysis report with insights on book and user patterns.
- Contribution to the field of recommendation systems and practical demonstration of ML applications in e-commerce and digital libraries.

PROJECT TIMELINE

Phase	Activity
Week 1–2	Data Collection & EDA
Week 3–4	Data Preprocessing & Feature Engineering
Week 5–6	Recommendation Algorithm Implementation
Week 7	FastAPI Backend Development
Week 8	React Frontend Development
Week 9	Integration & Testing
Week 10	Report & Submission

References (APA Style)

- Burke, R. (2002). Hybrid recommender systems: Survey and evaluation. *User modeling and user-adapted interaction*, 12(4), 331-370.
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- He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T. S. (2017). Neural collaborative filtering. In *Proceedings of the 26th international conference on world wide web* (pp. 173-182).
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- Ricci, F., Rokach, L., & Shapira, B. (2011). Introduction to recommender systems handbook. In *Recommender systems handbook* (pp. 1-35). Springer, Boston, MA.

DECLARATION

I hereby declare that the research work titled "Books Recommendation System" is original work carried out by me in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering at Lovely Professional University.

I affirm that this work has not been submitted elsewhere for any degree or diploma. All sources of information have been duly acknowledged, and the content reflects my independent research and development effort.

Signature

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Date: ____/____/____